

Heating

Flexible polyimide/silicone heater pads (Kapton heaters)

Radiant/ITO heated glass lid

wrapped heater band for the water bath

^^ big consideration is that i need to get around the plastic housing bad heating problem, idk if i can put strips of metal maybe so that i can have heaters on that

options looking at (vague initial ideas)

- **Chamber air heating** — forced convection: a small fan plus a resistive element (PTC heater or silicone rubber heater mat lining the walls), PID-controlled off a thermistor or PT100 RTD. This gets you overall enclosure temperature.
- **Plate/stage heating, independent of the air** — a heated insert directly under the 96-well plate (flexible Kapton/foil heater or a machined aluminum block with cartridge heaters), separately PID-controlled. This matters a lot for microscopy specifically: heating air alone creates thermal gradients that cause focus drift over a time-lapse, so touching the sample carrier directly with its own heater and sensor keeps both temperature and focus far more stable.
- **Lid heater** as the third, slightly-warmer zone, per above.

Top Glass Heating

^^ in an ideal world I'd order ITO glass but there are pretty much no south african supplier (sigma aldrich has but the sizing will not work)

^^ places like a abrisa technologies do this especially for incubators BUT it is overseas, and would take long to arrive and not clear that they'd do a single order sort of thing

^^MSE supplies also have some but the prices are around R3000-6000 which is crazy need to look at other options than ITO glass, it is the best solution but it is unfeasibly expensive

Other Heating Methods:

- Digkey: ITO coated PET, much cheaper: **R165**
https://www.digkey.co.za/en/products/detail/adafruit-industries-llc/1309/7241455?gclid=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLlg5ytnlC_OmD_SiQ-yiDdlzM&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qymKbK0mCfagDacfX76dkeylQfkYER34M9eWyrW5uwTWuBSqF0_Pn8aArRaEALw_wcB

^^ put htis film between two thin pieces of glass with copper tape bus and run voltage through for heating

^^ to make it better, the double glass is good because it helps improve the effects of outside temps.. less likely to drop temp rapidly and form condensation

^^ if the PET doesn't work, can heat the glass sides and have double glass layer (double glazed window vibe)

ITO Glass

Okay so after sourcing all of this above it was made clear to me that the glass is in budget so I'm going to source the ITO glass:

How It Works ?

The thermal performance relies on Joule heating:

$$Q = I^2 R t$$

The ITO coating acts as a microscopic resistive layer connected to busbar electrodes on opposite sides. When powered, electrical energy converts directly into surface heat. Integrating an external PID controller or adjusting input voltage ensures precise temperature regulation.

Key Features & Working Principle of the ITO Heating Glass

1. Four Core Advantages (<https://www.konshenglass.com/what-is-ito-heating-glass/>)
 - **High Transparency + Low Electrical Resistance:** Delivers over **80% visible light transmittance** while maintaining low resistivity (down to 10-4ohm.cm) , providing heat without sacrificing optical clarity.
 - **Uniform Heating & Precise Temperature Control:** Unlike standard heating wires that cause localized heating , ITO film heats the entire surface uniformly. Target temperatures can be precisely regulated between **30°C and 150°C**.
 - **Fast Response Rate:** Reaches target temperatures in seconds to a few minutes upon power application.
 - **Exceptional Weather Resistance:** Highly resistant to high temperatures and corrosion, making it reliable in demanding environments.

Sizing

Length and Width

96 well plate: 127.76 mm length, 85.48 mm width

^^ will need it to be reasonably bigger than this, if I want to put copper tape on for heating the glass

^^ I first looked at the "bus bar" for ITO glass just to see how much space they need - It will likely be 5-6mm wide, so need clearance for that and maybe some extra for space so let's say: Length: 128mm + 12mm + 15mm = 155mm, Width: 85mm +12mm+15mm = 112mm

Final: 155mm x 112mm

Thickness

Need to balance a few things here: cost, fragility, practical

so I'm going to see what the standard sizings are: 1.1mm from what I can see. For the larger glass size above I think that 1.1mm is ideal otherwise it is going to end up being too fragile

message to vendors: I'm sourcing ITO-coated glass for a microscope incubator project. I need a piece sized 150mm x 112mm, 1.1mm thick. Could you let me know the cost and lead time for this? I'm based in Cape Town, South Africa, so also keen to know about shipping to get a sense of total cost and timeline.

ohm/sq rating

Here's how it works: the resistance of your ITO coating between the two bus bars is R_s (ohms/sq) multiplied by the "number of squares" — which is just the distance the current travels (between your two bus bars) divided by the width of the bus bars themselves.

Then, since you know your voltage (12V), power comes from $P = V^2/R$.

For your panel (roughly 150mm between edges, 112mm wide), that number of squares works out to about 1.3. So plugging in a few common off-the-shelf sheet resistance values:

calc example:

$$R = R_s \times N = 10 \, \Omega/\text{sq} \times 1.34 = 13.4 \, \Omega$$

Using $P = V^2/R$:

$$P = 12^2 / 13.4 = 144 / 13.4 \approx 10.75 \, \text{W}$$

At 50 ohm/sq (same as the PET film you already looked at): around 2W — quite low, probably not enough heat on its own.

At 30 ohm/sq: around 3.5W — still modest.

At 10 ohm/sq: around 11W — a reasonable, meaningful amount of gentle heat for anti-condensation duty.

At 5 ohm/sq: around 21W — more than you likely need for just the lid.

power and temperature

$$\Delta T \text{ (temperature rise above room temp)} = P / (h \times A)$$

$$0.150\text{m} \times 0.112\text{m}, \text{ so } A \approx 0.0168 \, \text{m}^2$$

h is the convection coefficient - rough estimate here

$$\Delta T = 11\text{W} / (10 \times 0.0168) \approx 65^\circ\text{C above room temperature}$$

Options Vendors:

1. <https://www.konshenglass.com/contact/> (China) -> gonna go for this, lead time 10-15 days, they very helpful and responsive,
2. <https://www.uqgoptics.com/contact/?product-ref=ito-glass-windows-plates#contact-form> (UK)

3. <https://www.saidaglass.com/1mm-10ohm-clear-ito-conductive-glass-for-lab-testing.html> (china)

ITO Glass Electrode

Below are the options: (<https://www.konshenglass.com/what-is-ito-heating-glass/>)

To distribute electric current evenly across the ITO layer, busbar electrodes must be fabricated along the glass edges. Here are the primary electrode materials and methods:

Electrode Material / Process	Key Features & Advantages	Common Applications
Silver Paste (Screen Printing)	Most Popular Choice: Screen-printed and sintered/baked. Excellent conductivity, cost-effective, and strong adhesion.	Automotive side mirrors, freezer door glass, appliance panels.
Copper Foil / Copper Strip	Fixed via conductive adhesive or ultrasonic bonding. High current capacity and extremely low electrical resistance.	Large architectural glass panels, high-power heating windows.
Vacuum Evaporated / Sputtered Metals (Cr/Au/Ag/Ni)	Physical Vapor Deposition (PVD) metallic films. Ultra-thin, flat, and highly resistant to oxidation.	Precision optical equipment, high-end displays, defense gear.
Conductive Silver Adhesive / Aluminum Foil	Highly flexible and easy to apply.	R&D prototype development, specialized laminated structures.

Bus Bar Types: <https://abrisatechnologies.com/optical-coatings/bus-bars/>

- Chrome-Nickle-Gold – Solderable, Anti-Corrosion
- Silver Epoxy – Economical, Easy-to-Pattern, 3 Bond Surface Options, Wire Bonding
- Silver Frit – Solderable, Durable, High Adhesion
- Wire Soldering – Other Materials

I think Copper Strip is a good option because it's inexpensive and easy to source and comes as tape so it already sticks

Going to explore copper tape first:

- want it to be fairly thin (width) don't want to take up a lot of space on the glass

Options

1. DIY electronics: Copper Foil Tape 5mm 30meter, **R49**
<https://www.diyelectronics.co.za/store/connectors-wiring/1824-copper-foil-tape-5mm-30meters.html?>

[utm_campaign=google_shopping&utm_source=cpc&utm_medium=evergreen&utm_content=20484314766&gad_source=1&gad_campaignid=20484314766&gbraid=0AAADpm0I9d2KyS7NJ1Mf9lotletFctX&gclid=Cj0KCQjw16_UBhCqARIsAldOaXywEBF8KRhP6vaU6HwmLm4YTnGm2LTSI_V0FO0Xe8WbYc1qppKqoWlaAvKgEALw_wcB](https://za.rs-online.com/web/p/metallic-tapes/0285543?cm_mmc=ZA-PLA-DS3A--google--PLA_ZA_EN_PMAX_Catch+All--_285543&matchtype=&&gclid=Cj0KCQjw16_UBhCqARIsAldOaXywEBF8KRhP6vaU6HwmLm4YTnGm2LTSI_V0FO0Xe8WbYc1qppKqoWlaAvKgEALw_wcB)

2. RS Online: # RS PRO Conductive Metallic Tape, 6mm x 16.5 m 130 °C Max, **R385**, https://za.rs-online.com/web/p/metallic-tapes/0285543?cm_mmc=ZA-PLA-DS3A--google--PLA_ZA_EN_PMAX_Catch+All--_285543&matchtype=&&gclidsrc=aw.ds&gad_source=1&gad_campaignid=20574379828&gbraid=0AAAAADkeWNOlpeVHm_rzm8CDnY9KSQMeG&gclid=Cj0KCQjw16_UBhCqARIsAldOaXz7fH33a5fXtakdGN2bjPdnkD5XRKOejczhiDzqvGbttB9O6NQwbdEaAgsiEALw_wcB

3.

Bottom Glass Heating

So don't really want to add additional glass here because this part is super NB for image quality

^^ considering maybe a metal laser cut grid between wells that can be heated (on the top side)... ask Robyn about this

^^ can also just not do this at all... the rest of the incubator could very likely heat this well enough...

Chamber Heating

- Silicon Heater
^^ a lot of these are very large and would be awkward in the incubator
- Cartridge Heater
^^ these are nice because you can embed them in the incubator
- Heating Element

Options

1. RS Online: Heating Element, 10 W, 24V **R140**, https://za.rs-online.com/web/p/heating-elements/0336721?cm_mmc=ZA-PLA-DS3A--google--PLA_ZA_EN_PMAX_Catch+All--_336721&matchtype=&&gclidsrc=aw.ds&gad_source=1&gad_campaignid=20574379828&gbraid=0AAAAADkeWNNNoJ6I0Sv77j7wSax57oY86G&gclid=CjwKCAjwhZDUBhBG_EiwAbi5bjvUByl2ABhF5KrnHN8pKnGsSuZTrSi-f49AVkffv1TQDzKtGoxSI5hoCX_YQAvD_BwE
2. DIYelectronics: Heater Cartridge, 24V 40W, **R35**, https://www.diyelectronics.co.za/store/heater-cartridges/1395-heater-cartridge-6x20mm-24v-40w.html?utm_campaign=google_shopping&utm_source=cpc&utm_medium=evergreen&utm_content=20484314766&gad_source=1&gad_campaignid=20484314766&gbraid=0AAADpm0I-erQ_FJ7T78-

[eybBV4nS6nT&gclid=CjwKCAjwhZDUBhBGEiwAbi5bjnC9lnGNR9aHA-m_fG9VkfZcW5jUx7ojeVGQbIPStn1-jmuGeRQSOhoCRN8QAvD_BwE](https://www.bobshop.co.za/12pcs-20mmx140mm-film-heater-plate-adhesive-pad-pi-heating-elements-24v-72w-strip-heater-adhesive-po/p/690109290?gad_source=1&gad_campaignid=15305361599&gbraid=0AAAAADkeLXCtSN4humycQMA3yQ4iab0Fw&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qx_tSx--CHUQJN0Nvk8oOnMYVzobWqyfBceKYjxZugFsRYBQSwFQpMaAjsuEALw_wcB)

3. BobShop: Polyimide Heater, 24V 72W, **R518**, https://www.bobshop.co.za/12pcs-20mmx140mm-film-heater-plate-adhesive-pad-pi-heating-elements-24v-72w-strip-heater-adhesive-po/p/690109290?gad_source=1&gad_campaignid=15305361599&gbraid=0AAAAADkeLXCtSN4humycQMA3yQ4iab0Fw&gclid=Cj0KCQjw4orUBhCjARIsAlbF3qx_tSx--CHUQJN0Nvk8oOnMYVzobWqyfBceKYjxZugFsRYBQSwFQpMaAjsuEALw_wcB
4. Takealot: PTC Heater, 12V, **R427**, <https://www.takealot.com/5-pieces-12v-ptc-heater-70-c/PLID98344795>
5. Digikey, PTC Heater, 24V 160W, **R376**, https://www.digikey.co.za/en/products/detail/european-thermodynamics-ltd/HP06-2-10-12-24ET/22611974?gclsrc=aw.ds&gad_source=1&gad_campaignid=18332725675&gbraid=0AAAAADrbLijnZ_L0U6hsQhgUYgjeia_WQ&gclid=CjwKCAjwhZDUBhBGEiwAbi5bjuJKgVWH1ldYILCgZya_mepHSJt3XKviGMtlnKYpNI7uxatJiC3xxoCmwwQAvD_BwE
6. Amazon: # 20PCS Film Heater Plate Adhesive Pad 12V 12W 12V 13W,PI Heating Elements Film with Cord(12V 12W+12V 13W), **R570**, https://www.amazon.co.za/20PCS-Heater-Adhesive-Heating-Elements/dp/B0D9BF5HQH/ref=sr_1_2?dib=eyJ2IjojMSJ9.J252LE6x7ClmvR6PW9NU3QHym6yn5YPxlk5cMq2G9SajR1r5MJ307-Yh55td5fBjevnM44F293OVUV6J8DRTCcHuY672u4xEQHdDjER5_6HLkIMzOCm1vyCXlekXeAld.DduSQ1tjDI24erwzsJacfKJSv_F5EqzHGtGMdCRPalg&dib_tag=se&keywords=heater+polyimide&qid=1787048458&sr=8-2

on most of the incubators i see they used the polyimide heaters, the amazon looks the best